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INTRODUCTION TO PROGRAMMING PARADIGM

Overview of programming paradigms:

What is programming paradigms?

A programming paradigm is a fundamental style or approach to programming that guides how solutions are designed and implemented.

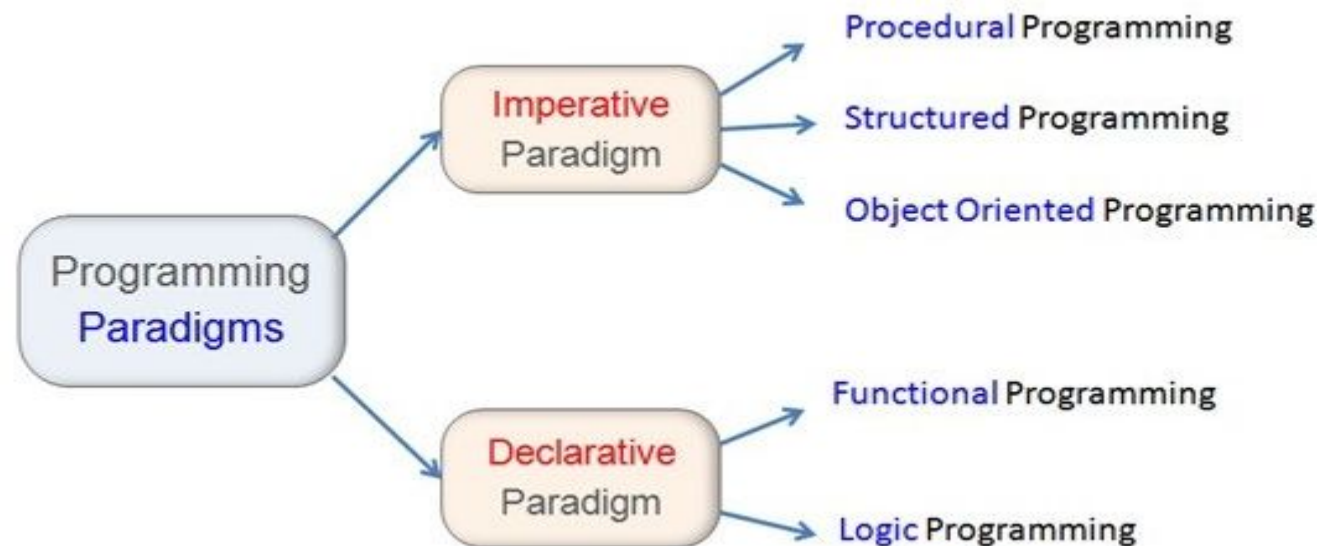
Purpose: Provides a structured way to organize code and solve problems effectively.

Real-life Example

An **online food ordering system** can demonstrate various programming paradigms: procedural programming handles order processing step-by-step, object-oriented programming models entities like customers and orders, functional programming applies pure functions for calculations, logic programming uses rules for recommendations, and scripting automates notifications and updates.

Main types in programming paradigms

- **Procedural Programming** – Step-by-step instructions using functions and procedures.
- **Object-Oriented Programming (OOP)** – Organizes code around objects and data.
- **Functional Programming** – Focuses on pure functions and immutability.
- **Logic Programming** – Uses facts and rules to deduce new information.
- **Scripting** – Lightweight, interpreted code for automation and web tasks.



Procedural programming:

What is procedural programming ?

A programming paradigm based on the concept of procedure calls, where programs are structured as a sequence of instructions grouped into functions or procedures.

Key Characteristics: Step-by-step task execution using variables, loops, and conditionals, with modular code through reusable procedures/functions.

Why Procedural Programming?

- **Simplicity:** Easy to understand and implement for beginners
- **Structured Approach:** Organizes code into clear, logical steps
- **Reusability:** Functions/procedures can be reused in multiple parts of the program
- **Efficiency:** Produces fast-executing code for small to medium-sized tasks
- **Widespread Use:** Supported by many popular languages like C, Pascal, and Python

Object-Oriented Programming:

Definition:

A programming paradigm that organizes code around objects—data structures containing attributes (data) and methods (behavior)—to model real-world entities.

Key Characteristics:

- Encapsulation: Bundling data and methods together
- Inheritance: Reusing properties and behaviors from existing classes
- Polymorphism: Using a single interface for different underlying forms
- Abstraction: Hiding implementation details and showing only essential features

Why Object-Oriented Programming?

- Better code organization and maintainability
- Encourages reuse through inheritance and modular design
- Ideal for large, complex applications like ERP systems, banking software, and e-commerce platforms

Functional programming:

Definition:

A programming paradigm where computation is treated as the evaluation of mathematical functions, avoiding changes to state and mutable data.

Key Characteristics:

- **Pure Functions:** Always produce the same output for the same input, without side effects
- **Immutability:** Data cannot be modified after creation
- **Higher-Order Functions:** Functions can accept other functions as arguments or return them
- **Function Composition:** Combining simple functions to build more complex ones

Why Functional Programming?

- Reduces bugs by avoiding shared state and side effects
- Simplifies reasoning about code
- Ideal for data processing, AI, and distributed systems

Logic Programming:

Definition:

A programming paradigm where facts and rules are defined, and the program uses logical inference to derive conclusions or answer queries.

Key Characteristics:

- Uses facts, rules, and queries
- Declarative style—focus on *what* to solve, not *how*
- Relies on an inference engine for reasoning
- Strong in pattern matching and knowledge representation

Advantages:

Provides high-level abstraction for problem-solving and is easy to extend without changing flow. Ideal for AI, expert systems, pattern matching, and complex search problems.

Why Logic Programming:

- **Declarative Approach:** Focus on defining the problem, not the solution steps.
- **Knowledge Representation:** Ideal for storing and reasoning over facts and rules.
- **Flexibility:** Easily update or extend the knowledge base without rewriting logic flow.
- **AI and Expert Systems:** Naturally suited for inference-based problem solving.
- **Complex Query Handling:** Efficient for pattern matching, relationship finding, and constraint solving.

Comparison of Different Programming Paradigms:

Aspect	Procedural	Object-Oriented (OOP)	Functional	Logic
Approach	Step-by-step instructions	Organizes code into objects with data & behavior	Focus on pure functions & immutability	Define facts & rules, solve via inference
State Management	Uses variables & modifies state	Encapsulated state within objects	Avoids shared mutable state	No explicit state, reasoning based on facts
Reusability	Functions/procedures	Classes & inheritance	Higher-order functions & composition	Reusable rules/facts
Best Suited For	Small to medium tasks, procedural workflows	Large, complex systems modeling real-world entities	Data transformation, parallel processing	AI, expert systems, complex queries
Examples	C, Pascal	Java, C++, Python (OOP)	Haskell, Scala, Java (functional features)	Prolog, Datalog

Scripting Programming – Characteristics of Scripting Languages

- **Interpreted Execution:** Typically executed by an interpreter rather than compiled to machine code.
- **Rapid Development:** Minimal setup, easy to write and modify code quickly.
- **Dynamic Typing:** Variable types determined at runtime, offering flexibility.
- **High-Level Abstraction:** Provides built-in support for common operations (file handling, string manipulation, etc.).
- **Automation Friendly:** Often used for automating repetitive tasks.
- **Integration Capability:** Easily integrates with other programs, APIs, and system tools.
- **Platform Independence:** Scripts can often run on multiple OS environments without modification.

Scripting vs Traditional Programming Languages:

Feature	Scripting Languages	Traditional Programming Languages
Execution	Interpreted at runtime	Compiled to machine code
Development Speed	Faster to write and modify	Slower, requires compilation steps
Typing	Usually dynamically typed	Usually statically typed
Performance	Slower due to interpretation	Faster due to compilation
Use Case	Automation, web scripts, quick tasks	Large-scale, performance-critical apps
Examples	Python, JavaScript, Bash	C, C++, Java

Advantages of Scripting Languages:

- **Rapid Development:** No compilation step; faster prototyping.
- **Ease of Learning:** Simple syntax and minimal setup.
- **Flexibility:** Works well across platforms and integrates with other tools.
- **Automation Capabilities:** Perfect for repetitive or batch processes.
- **High-Level Features:** Built-in libraries for networking, file I/O, data parsing, etc.

Use Cases of Scripting Languages:

- **Web Development:** JavaScript for client-side interactivity, PHP/Python for server-side logic.
- **System Administration:** Bash or PowerShell for automating OS tasks.
- **Data Analysis:** Python or R for processing and visualizing data.
- **Game Development:** Lua for game scripting.
- **Automation & Integration:** Python scripts for connecting APIs, generating reports, or running batch jobs.

Overview of popular scripting languages: Python, JavaScript, Bash - Introduction to HTML and CSS:

1. Python

- **Type:** High-level, interpreted, dynamically typed
- **Strengths:** Easy syntax, extensive libraries, cross-platform
- **Use Cases:** Data analysis, web development (Django/Flask), automation, AI/ML
- **Example:** Automating file renaming, building REST APIs, data visualization

2. JavaScript

- **Type:** Interpreted, dynamic, event-driven
- **Strengths:** Runs in browsers, supports both frontend & backend (Node.js)
- **Use Cases:** Web interactivity, single-page apps, server-side scripting
- **Example:** Interactive forms, real-time chat apps, animations on websites

Popular Scripting Languages:

3. Bash

- **Type:** Unix shell scripting language
- **Strengths:** Native to Linux/Unix environments, excellent for system automation
- **Use Cases:** Automating backups, running batch jobs, system monitoring
- **Example:** Script to back up log files daily to a server

Introduction to HTML :

HTML (HyperText Markup Language)

- **Purpose:** Structure and content of web pages
- **Features:** Defines elements like headings, paragraphs, links, images, forms

Example:

```
<h1>Welcome</h1>
```

```
<p>This is a paragraph.</p>
```

Introduction to CSS:

Definition

A style sheet language used to control the appearance and layout of HTML elements on a web page.

Purpose: Separates website content (HTML) from its presentation (design).

What it Does: Defines colors, fonts, spacing, sizes, backgrounds, animations, and positioning of elements

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Features:

- Controls style and layout of web pages
- Manages colors, fonts, spacing, and responsiveness
- Ensures visual consistency across multiple pages

Benefit: Makes websites visually appealing, responsive, and easy to maintain.

Example:

```
h1 {  
  color: navy;  
  text-align: center;  
}
```